

Session 5 Assignment Problems – Nested if Statements and Compound Relational Conditions

For each problem, develop the IPO and Code.

1. The input to the problem is quantity of widgets. Your program should determine the price to charge based on the schedule below. Calculate the extended price (quantity x price). Calculate tax at 7%. Display the extended price, tax amount and total.

Quantity Price

>10000 \$10

5000 to 10000 \$20

Below 5000 \$30

Input	Process	Output
Qty of Widgets	If Qtny > 1000: Wprice = 10 Else if (Elif) Qtny >= 5000 Wprice= 20 Else: Wprice = 30	Extended Price
	Extended Price = Qty * Price	Tax
	Tax = Extended Price * 7%	Total
	Total = Extended Price + Tax	

2. Enter a part number and quantity Determine the cost per unit using the table below. Then calculate the total cost (quantity x unit cost). Display the part number, cost per unit and total cost. Note: Part number can be an integer but it can also be a string because you are not doing arithmetic on it. However in your code if statement be sure to compare using consistency, that is, if item == "10" when item is a string and if item == 10 when item is an integer.

Part Unit Cost

10 or 55 1.00

99 2.00

80 or 70 3.00

All others 5.00

Input	Process	Output
Part Number	Total Cost = Qtny * Unit Cost	Part Number
Qtny	If part == 10 or 55 Unit Price = \$1 Elif: == 99 Unit Price = \$2 Elif; ==80 or 70 Unit Price = \$3 Else: Unit Price \$5	Cost per Unit
	Total Cost = Qtny * UnitPrice	Total Cost

3. Enter a principle amount of a CD and year to maturity of CD. Determine the interest rate based on the amount of the principle and maturity (see below). Calculate first year interest (principle x interest rate). Display principle, interest rate and the interest amount for first year.

Principle Years to Maturity Interest Rate

>\$100,000 5 6%

\$50,000 to \$100,000 10 5%

\$50,000 to \$100,000 5 4%

Any other principle and years 2%

Input	Process	Output
Principle Amount of a CD	1 YR IntRate = Principle Amount * Intrest Rate	Principle Amount
Years into CD Maturity	If Principle Amount > 100000 and 5yrs Intrest Rate = 6% Elif 50000 <= Principle Rate <= 100000 and 10yrs Intrest Rate = 5% Elif 50000 <= Principle Rate <= 100000 and 5yrs Else: Interest Rate = 2%	Intrest Rate
		Intrest Amount for 1 YR

4. Allow the user to enter number of concert tickets. The price per ticket depends on the volume (see below). Display the number of tickets, price per ticket and the total cost (number of tickets x Price Per Ticket).

Quantity Price Per Ticket

>=25 \$50

10 to 24 \$60

5 to 9 \$70

Less 5 \$75

Input	Process	Output
Number of Concert Tickets	If NumberofConcertTickets >= 25: Price per Ticket = \$50 Elif 10 <= NumberofConcertTickets <= 24 Price per Ticket = \$60 Elif 5 <= NumberofConcerTickets < =9: Price per Ticket = 70 Else: Price per Ticket = 75	Number of Tickets
		Price per Ticket
	Total Cost = Number of Tickets * Price per Ticket	Total Cost

5. The user will enter employee last name, salary and job level (as noted below). Use the job level to determine the bonus rate. Then compute bonus to be salary times bonus rate. Display employee last name and bonus.

Job Level Bonus Rate

10 and above 25%

5 to 9 20%

All others 10%

Input	Process	Output
Last Name	Bonus = Salary * Bonus Rate (%)	Last Name
Salary	If Job Level >= 10: BonusRate = 25% Elif 5 <= Job Level <= 9: BonusRate = 20% Else: BonusRate = 10%	Bonus
Job Level		